# Ask the Regulations: A Type-Stratified Study of Hybrid Retrieval over Small Domain Corpora with Evidence-Audited Ontologies

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**Abstract.** Retrieval-augmented systems deployed in regulated domains face a reality rarely discussed in benchmark papers: the corpus is small and grows organically, queries fall into sharply different types, and every answer must be traceable to authoritative text. We study this regime on a registered corpus of 27 assets—20 public marine-dumping regulatory documents from the national environmental authority plus 7 operational assets. We first construct a domain ontology (100 nodes, 200 relations) in which every LLM-extracted node and relation must cite its supporting passage, and we audit the evidence chain with deterministic string verification: 96% of node names and 85% of relations are fully traceable to the cited source text, and 89.9% of citations verify verbatim. Second, we build a 306-question type-stratified benchmark (title-lookup, entity/content, multi-hop, refusal) with deterministically constructed gold answers. Third, we compare six retrieval configurations with per-question paired Wilcoxon tests and find that *no single configuration wins across query types*: plain BM25 dominates title queries (hit@1 .901); graph-expanded lexical retrieval dominates entity and multi-hop queries (hit@5 .930/.983); and a strong dense retriever with a cross-encoder reranker—often assumed superior—is significantly worst on *every* type on this corpus of near-duplicate official documents. A deterministic two-system router that sends title-coverage queries to BM25 and everything else to the graph retriever captures the per-type oracle: it significantly beats every single configuration (all $p < 0.005$) and is statistically indistinguishable from the per-type upper bound ($p = 0.10$). For abstention, no configuration's top-1 scores separate answerable from out-of-domain queries (zero-floor overlap, inversion, and low-tail overlap, respectively), cautioning against score-based refusal in small-corpus deployments. We release the benchmark, audit tooling, router, and all configurations.

**Keywords:** Domain retrieval · Hybrid retrieval · Adaptive retrieval · Knowledge ontology · Evidence audit · Retrieval evaluation · Abstention

## 1 Introduction

Most retrieval research is evaluated on large, heterogeneous corpora where dense retrieval and cross-encoder reranking are known to excel [5, 2]. The deployment reality in regulated domains—environmental compliance, permits, tech-

nical standards—is almost the opposite. The corpus is small (tens of documents, not millions) and grows one announcement at a time; documents are near-duplicates of each other, sharing boilerplate and titles that differ by a single edition number; queries fall into sharply different types, from title lookup to multi-hop reasoning across related instruments; and, because a wrong citation can mislead a real governance decision, every answer must be traceable to authoritative text. Small-corpus retrieval is not merely large-corpus retrieval with fewer documents: score distributions, ranking granularity, and the behaviour of fusion strategies all change.

This paper reports a controlled, type-stratified study of retrieval in exactly this regime, on a real deployment: a knowledge service that backs a marine debris governance agent for aquaculture zones. The service ingests public regulatory documents from the national environmental authority (marine dumping area announcements, standards, permit rules, sewage outfall guidelines, and comprehensive governance plans), maintains a domain ontology, and answers queries with citation-backed evidence.

We address three research questions:

**RQ1.** On a small, near-duplicate regulatory corpus, how do lexical, graph-expanded, dense, and hybrid retrieval configurations compare *per query type*, with paired significance tests?

**RQ2.** Can an LLM-extracted domain ontology be made evidence-traceable by construction, and audited deterministically without a gold standard?

**RQ3.** Is score-threshold abstention (refusing out-of-domain queries) viable, and is any threshold portable across configurations?

*Contributions.*

1. **An evidence-audit framework for LLM-extracted ontologies**, Sect. 3: every node and relation must cite a passage of a registered asset; deterministic verbatim verification reports citation verifiability (89.9%, 302/336), relation-endpoint traceability (90.0%), full relation traceability (85.0%), and node-name traceability (96%). To our knowledge this audit protocol—verify every fact by exact string match against its cited source, with no model in the loop—is reusable on its own.
2. **A type-stratified 306-question benchmark** (Sect. 4) with four query types (title, entity/content, multi-hop, refusal) and deterministic gold construction: zero model calls, fixed templates, gold taken from the ontology and asset tables themselves. The corpus is a live deployment's and grew organically during the study (15→27 assets); we discuss what that growth does to per-type conclusions in Sect. 6.
3. **A controlled comparison of six configurations with paired significance tests** (Sect. 6): the headline finding is that *no configuration wins across types*—BM25 owns title queries, the graph-expanded lexical retriever owns entity and multi-hop queries, and a strong dense-plus-reranker pipeline is significantly worst on all three types; several common assumptions (dense-plus-reranker dominance, fusion as a free lunch) do not survive.

4. **A two-system rule router that matches the per-type oracle** (Sect. 7): a deterministic title-coverage signal routes queries between BM25 and the graph retriever; it significantly beats every single configuration and is statistically indistinguishable from the per-type upper bound, so adaptive retrieval in this regime needs two systems and one cheap signal, not a learned router.
5. **An abstention fragility result** (Sect. 7): no configuration's top-1 scores separate answerable from refusal queries (zero-floor overlap, inversion, low-tail overlap), arguing for evidence-coverage-based refusal.
6. **Released artifacts**: benchmark, all six configurations and the router, the audit and significance tooling.[1]

## 2 Related Work

*Hybrid retrieval.* Lexical (BM25 [1]) and dense retrieval [2, 3], refined by cross-encoder rerankers [16], are commonly fused; reciprocal rank fusion (RRF) [4] is a robust parameter-light combiner. BEIR [5] showed that no single retriever dominates across heterogeneous corpora; we show the same holds *within* one small domain across *query types*, a granularity usually hidden by corpus-level averages. Domain benchmarks such as LegalBench [21] measure professional reasoning at scale; our benchmark complements them with a within-deployment, type-stratified study of the retrieval layer itself.

*Adaptive and routed retrieval.* Adaptive-RAG [17] trains a complexity classifier that routes each query to no-, single-, or multi-step retrieval; Self-Route [18] lets the model itself decide between retrieval and long-context reading. Both target large, heterogeneous corpora and pay for a learned or prompted router. We show that in the small-corpus regime the router can be had for free: a single deterministic title-coverage statistic captures the per-type oracle.

*KG-augmented retrieval.* GraphRAG [6], LightRAG [19], and the HippoRAG line [7, 20] augment retrieval with graph structures extracted from text, mostly over large corpora. We study the complementary regime: a small, curated, versioned ontology where graph expansion is a precision instrument for multi-hop queries—and show it *hurts* title queries relative to plain BM25.

*LLM knowledge construction and verifiability.* LLMs are now used to construct knowledge graphs [8, 9], but extraction hallucination is a documented failure mode [10]. Our audit is a deterministic, model-free check that every extracted fact carries a verbatim-verifiable citation—related in spirit to grounding checks for RAG [11] but applied to ontology construction, where the unit of verification is a triple plus its cited evidence.

[1] Anonymized repository: `https://anonymous.4open.science/r/ask-the-regulations` (placeholder to be created before submission).

*Abstention in retrieval and RAG.* Calibration and knowing-when-not- to-answer are studied for LLMs [12, 13] and for retrieval-augmented pipelines such as Self-RAG [14]. We contribute a concrete negative result: on a small regulatory corpus, top-1 retrieval scores do not yield a portable abstention threshold, and their failure mode differs per configuration (zero-floor overlap, inversion, or low-tail overlap).

## 3 Corpus and Evidence-Audited Ontology

### 3.1 Corpus

The evaluation corpus mirrors a production knowledge service for marine debris governance. It contains **27 assets**: 20 public regulatory documents harvested from two official columns of the national environmental authority (marine dumping-area announcements, national standards and their draft-for-comment notices, permit rules, sewage outfall technical guidelines, and a regional comprehensive governance plan), plus 7 operational demo assets (event records, telemetry, monitoring datasets) that the service also indexes. Each asset is registered once and versioned immutably.

### 3.2 Ontology construction with mandatory citations

The ontology (100 nodes, 200 relations over the current published version) is built through a candidate-extraction, human-review, publish lifecycle. The extraction layer mixes rule-based extraction (167 relations) and LLM extraction (33 relations). The invariant that makes the ontology auditable is *mandatory evidence*: every node and every relation must carry at least one evidence item $\langle$asset_id, asset_version_id, citation$\rangle$, where *citation* is a snippet of the cited asset's text. Ontology publishing records the version id, so downstream decisions can pin an ontology version.

### 3.3 Deterministic evidence audit

The audit verifies, per relation: (i) each cited snippet appears *verbatim* (normalized whitespace) in the cited asset's registered text; and (ii) both endpoint names occur in the text of a successfully verified citing document. Per node: the node name (or an alias) occurs in its source document. All checks are exact string operations over the same registered corpus—no model calls, no manual annotation claims. We deliberately call this an *evidence-traceability audit*, not a gold-standard precision/recall evaluation: a fully traceable relation can still be domain-silly, and recall of "all true facts" is undefined without expert annotation.

Table 1 shows the audit outcome. The 11% non-verbatim citations are concentrated in LLM-extracted relations whose snippets paraphrase rather than quote; they are flagged, not silently accepted. The untraceable sample is logged with reasons (*citation-not-in-doc*, *endpoint-missing-in-citing-doc*) and feeds the human review queue.

**Table 1.** Evidence audit of the published ontology (27 assets, 100 nodes, 200 relations).

| Audit metric | Result | Rate |
|---|---|---|
| Citations verbatim-verifiable | 302/336 | 89.9% |
| Relation endpoints traceable | 180/200 | 90.0% |
| Relations fully traceable (citations + endpoints) | 170/200 | 85.0% |
| Node names traceable to source doc | 96/100 | 96.0% |
| Extraction method | 167 rule / 33 LLM | — |

**Table 2.** Benchmark composition. Gold answers are constructed deterministically from the ontology and asset tables; zero model calls.

| Type | $n$ | Gold Construction |
|---|---|---|
| Title lookup (single) | 81 | 1 asset 3 fixed templates over asset titles |
| Entity/content (single2) | 100 | 1 asset 2 fixed templates per reviewed ontology entity; no title words |
| Multi-hop | 120 | source assets of chain 2-hop chains $A \to B \to C$ from published relations |
| Refusal | 5 | none out-of-domain questions |

## 4 A Type-Stratified Benchmark

The benchmark has 306 questions in four types (Table 2), motivated by the traffic a governance knowledge service actually sees: *title lookup* ("what does the notice on X require?"), where the title is the strongest lexical signal; *entity/content queries*, which name an entity or requirement but never use title words, forcing content-level matching; *multi-hop queries* derived from two-hop relation chains, whose gold assets are the sources of the chain entities—no single document mentions both endpoints; and *refusal queries* (out-of-domain), for which the correct behaviour is to abstain.

Two design choices matter for reproducibility. First, templates are fixed and gold answers are read from published ontology/asset tables, so the benchmark is regenerable deterministically. Second, the corpus grew organically during system development (15→27 assets); we evaluate the 27-asset version throughout. Table 3 shows one real question per type (translated from Chinese).

## 5 Experimental Setup

All configurations share the corpus representation: assets are chunked at 800 characters, each chunk prefixed with the asset title (capped at 1600 characters). Table 4 summarizes the six systems.

BM25 uses our own index (deterministic tokenizer, no segmenter dependency); Lex+Graph is the deployed production service; Dense+Rerank and the hybrids run in the evaluation harness over identical chunks.

**Table 3.** One real question per type (translated from Chinese; asset ids verbatim).

| Type | Question | Gold assets |
|---|---|---|
| Title | What are the governance requirements in the notice soliciting comments on the national standard *Online Monitoring of Ocean Dumping* (draft)? | OD-SEA-EX-12 |
| Entity | What are the management requirements for the "five dumping areas"? Retrieve the supporting documents. | OD-SEA-EX-03 |
| Multi-hop | How is the Binzhou north offshore dumping area connected to the responsible organization in the governance process? | EX-02, EX-09, STD-2025-01 |
| Refusal | What subsidy policies exist for island tourism homestays? | (none—abstain) |

*Metrics.* We report hit@{1,3,5} and MRR over the ranked asset list of each system; refusal queries carry no gold asset and are analysed separately in Sect. 7. All significance claims use two-sided paired Wilcoxon signed-rank tests over per-question reciprocal rank, with Holm correction within families (Sect. 7).

*Implementation details.* The harness is pure Python. BGE-M3 embeddings and BGE-Reranker-V2-M3 relevance scores are obtained from a managed inference endpoint and cached, so every number regenerates from the frozen artifact without further model calls; the lexical+graph baseline is the production service queried over HTTP on the same machine. All model calls are cached; every number in this paper is recomputable from the frozen run artifact.

## 6 Results

*F1: No configuration wins across query types.* Table 5 is the core result, and the specialisation is sharp. Plain BM25 owns title queries: hit@1 .901 and hit@5 1.000, significantly above every other system (vs. Lex+Graph $p$=4.8×10$^{-8}$, vs. RRF Hybrid $p$=9.0×10$^{-4}$)—title matching is exactly what BM25's term weighting does. The graph-expanded lexical retriever owns the other two types: entity hit@5 .930 (vs. BM25 $p$=1.3×10$^{-2}$) and multi-hop hit@5 .983 (vs. BM25 $p$=1.6×10$^{-3}$)—when the query names entities rather than titles, graph expansion supplies the vocabulary bridge. Each specialist loses where the other wins (Lex+Graph drops to .519 hit@1 on titles; BM25 drops to .840 and .950 hit@5 on entity and multi-hop), and the deltas are significant on both sides.

**Table 4.** The six configurations.

| Name | Description |
|---|---|
| BM25 | BM25 over chunks ($k_1$=1.5, $b$=0.75); CJK text tokenized as character bigrams, ASCII runs as whole tokens; no graph expansion. |
| Lex+Graph | Production retriever: weighted lexical scoring (title ×6, body ×1.5, standard-number ×2) plus one-hop ontology graph expansion. |
| Dense+Rerank | BGE-M3 embeddings; cosine top-30 chunks; BGE-Reranker-V2-M3 reranks top-20 chunks; asset rank = its best chunk. |
| RRF Hybrid | RRF ($k$=60) fusion of Lex+Graph asset ranking and dense asset ranking. |
| TB Hybrid | RRF hybrid whose dense channel embeds title-boosted chunks (title repeated 3×). |
| Rule router | Deterministic title-coverage signal (character-bigram overlap $\geq$0.5 with any registered asset title): title queries → BM25, everything else → Lex+Graph. |

*F2: Dense retrieval with cross-encoder reranking is significantly worst on every type.* The configuration most often recommended by default— bi-encoder retrieval plus a cross-encoder—loses significantly on all three types: title $p$=.014, entity $p<10^{-6}$, multi-hop $p<10^{-7}$, each vs. Lex+Graph. Its title hit@1, hit@3 and hit@5 are all .481: the gold document is either ranked first or missing from the reranker's candidate window entirely. The mechanism is structural: this corpus contains clusters of near-duplicate official documents; whole-chunk embeddings dilute the few discriminative tokens (an edition year, a place name), and the reranker can only reorder the dense top-20 chunks. Dense retrieval's known weakness on near-duplicates [5] dominates when the *entire corpus* is near-duplicates of a few families.

*F3: Fusion is a strong single system, not a free lunch.* RRF fusion of the graph retriever with the dense channel lifts title queries far above graph-only retrieval (hit@1 .519→.815, $p$=1.1×10$^{-7}$) while preserving multi-hop (.983 hit@5), which makes it the best *single* deployable configuration. But it remains significantly below the title specialist (BM25 .901 hit@1, $p$=9.0×10$^{-4}$) and below the graph specialist on entity queries (.860 vs. .930 hit@5, $p$=4.0×10$^{-3}$); title-boosting the dense channel (TB Hybrid) adds little over plain RRF. Fusion averages toward the specialists instead of reaching them—the gap is exactly what routing should recover.

*F4: The deployment regime ages benchmarks.* The corpus grew organically during system development (15→27 assets, the added documents being highly similar dumping/outfall announcements), and every absolute number moved with it. Corpus and benchmark changed together, so we treat growth as a property

**Table 5.** Main results on the 306-question benchmark (frozen run). Per type: hit@1 / hit@5 / MRR over ranked assets. Bold = significantly best in the column (paired Wilcoxon, $p < 0.05$, see Table 6).

| | Title ($n$=81) | | | Entity ($n$=100) | | | Multi-hop ($n$=120) | | |
|---|---|---|---|---|---|---|---|---|---|
| Config | h@1 | h@5 | MRR | h@1 | h@5 | MRR | h@1 | h@5 | MRR |
| BM25 | **.901** | **1.000** | **.943** | .510 | .840 | .642 | .692 | .950 | .791 |
| Lex+Graph | .519 | .852 | .647 | **.600** | **.930** | **.740** | .800 | **.983** | .890 |
| Dense+Rerank | .481 | .481 | .481 | .370 | .450 | .410 | .575 | .725 | .631 |
| RRF Hybrid | .815 | **1.000** | .882 | .510 | .860 | .650 | .783 | **.983** | .870 |
| TB Hybrid | .778 | **1.000** | .859 | .530 | .870 | .668 | .792 | **.983** | .880 |
| Rule router | .840 | **1.000** | .913 | **.600** | **.930** | **.740** | **.808** | **.983** | **.894** |

**Table 6.** Selected paired Wilcoxon signed-rank tests over per-question reciprocal rank (two-sided; zeros dropped). Full matrix in the artifact.

| Type | Pair | $p$ | Reading |
|---|---|---|---|
| Title | BM25 vs. Lex+Graph | $4.8\times10^{-8}$ | BM25 owns title lookups |
| Title | RRF vs. Lex+Graph | $1.1\times10^{-7}$ | fusion lifts titles |
| Entity | Lex+Graph vs. Dense | $1.5\times10^{-7}$ | dense worst on content |
| Entity | Lex+Graph vs. BM25 | $1.3\times10^{-2}$ | graph pays on content |
| Multi-hop | Lex+Graph vs. Dense | $7.7\times10^{-8}$ | dense worst on multi-hop |
| Multi-hop | Lex+Graph vs. BM25 | $1.6\times10^{-3}$ | graph bridges +3.3pt |

of the regime rather than a controlled variable—and type-stratified reporting is what keeps such drift visible instead of hiding it in an average.

## 7 A Two-System Router and Abstention

*The rule router captures the oracle.* The oracle that routes each query type to its best system (title → BM25, entity and multi-hop → Lex+Graph) reaches overall hit@5 .970 / MRR .855. The rule router approximates it with one deterministic signal—character-bigram overlap ≥0.5 between the query and any registered asset title—because title coverage is exactly what separates the type BM25 wins from the types graph expansion wins. It routes 73/81 title queries to BM25 and sends everything else to the graph retriever (13 non-title queries are misrouted to BM25); overall routing accuracy is 93.0% (280/301 answerable queries). The full routing confusion is compact (Table 7): errors concentrate on multi-hop queries whose text happens to quote an asset title. The result (Table 5, last row): overall hit@5 .970 / MRR .848, with per-type numbers at or above each specialist's own (multi-hop hit@1 .808, slightly above Lex+Graph's .800—some misroutes happen to help). Paired tests show the router *significantly beats every single configuration* (vs. Lex+Graph $p$=$2.3\times10^{-7}$, vs. BM25 $p$=$3.3\times10^{-4}$, vs. both hybrids $p$<$5\times10^{-4}$) and is statistically indistinguishable from the per-type

**Table 7.** Rule-router confusion (answerable queries): routing decision by query type. Routing is correct when title queries go to BM25 and the rest to the graph retriever: 280/301 (93.0%).

| Query type | → BM25 | → Lex+Graph |
|---|---|---|
| Title (81) | 73 | 8 |
| Entity (100) | 5 | 95 |
| Multi-hop (120) | 8 | 112 |
| Routing-correct | 280/301 (93.0%) | |

**Table 8.** Abstention: top-1 retrieval scores of answerable vs. refusal queries. Three behaviours across configurations.

| Config | min answerable | max refusal | behaviour |
|---|---|---|---|
| Lex+Graph | 0.0 | 46.5 | zero-floor overlap |
| BM25 | 7.37 | 13.72 | *inverted* (refusal scores higher) |
| Dense+Rerank | 0.0008 | 0.0031 | low-tail overlap |

oracle ($p$=0.10). This mirrors the routing philosophy of Adaptive-RAG [17] with a regime- appropriate simplification: where large heterogeneous corpora need a trained complexity classifier, here one deterministic string statistic suffices. All highlighted comparisons in Tables 6 and 7 survive Holm correction within their test families (largest adjusted $p$ .013 for the specialist contrasts; $9.9\times10^{-4}$ for the router family). A post-hoc threshold sweep (0.1–0.9; artifact) shows the router is not knife-edge: overall hit@5 stays within .95–.97 for any cutoff in $[0.2, 0.5]$. The sweep also reveals that a lower cutoff (0.2, routing 194/301 queries to BM25) would yield slightly higher MRR than our a-priori 0.5 (+.023, paired $p$=.022)—we keep the a-priori value to avoid post-hoc tuning and recommend a per-domain sweep in practice. The practical reading for small-corpus deployments: adaptive retrieval here needs two indexes and one string-overlap signal—not a learned router, not fusion.

*Abstention: no configuration separates.* With only 5 refusal queries this is a case study, not a powered experiment—but the pattern is consistent and stark (Table 8, Fig. 1). Under Lex+Graph, four answerable queries return *no lexical hit at all* (top-1 score exactly 0) while refusal queries score up to 46.5: the answerable low tail reaches the floor of the scale, far below the refusals—a zero-floor overlap. Under BM25 the boundary *inverts*: one out-of-domain question (13.7) outscores the weakest answerable questions (min 7.4)—BM25 rewards long out-of-domain queries that match high-IDF bigrams. Under dense scores the refusal cluster ($1.1–3.1\times10^{-3}$) sits inside the answerable low tail (min $8\times10^{-4}$). No cutoff separates answerable from refusal under *any* configuration, and a threshold tuned on one configuration transfers to none. Our production QA layer therefore ab-

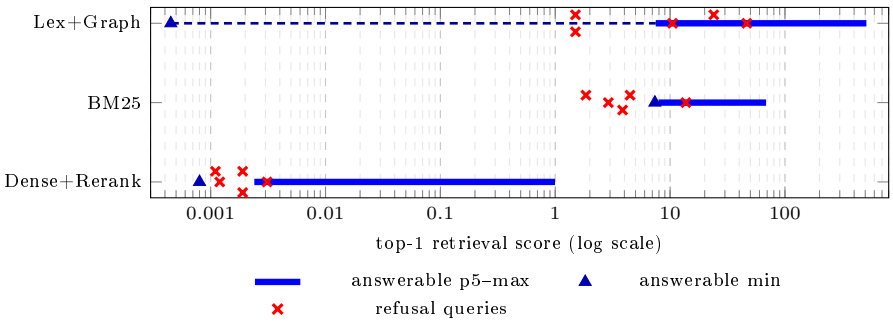



**Fig. 1.** Top-1 score ranges for answerable vs. refusal queries per configuration (log scale; identical data as Table 8; the band spans the 5th–100th percentile of answerable scores; coincident refusal marks within a row are offset vertically for visibility). Under Lex+Graph, four answerable queries score exactly 0 (marker at the scale floor, dotted leader to the band) and refusal marks reach deep into the band's low range; under BM25 one refusal mark lands inside the band (inversion); under Dense+Rerank the refusal cluster overlaps the band's left tail. No configuration separates the two populations.

stains on *evidence coverage* (are the retrieved citations sufficient to ground an answer?) rather than retrieval score, and states its refusal to the user.

## 8 Discussion and Limitations

*Threats to validity. Construct:* benchmark questions are generated from the same tables the retrievers index—title queries quote titles (favouring BM25), entity and multi-hop queries carry ontology vocabulary (favouring graph expansion). This mirrors real governance traffic, where users cite instrument titles and entity names—which is exactly the signal the router exploits—but it also means absolute numbers are protocol-relative and must not be compared with open-domain benchmarks. *Internal:* all contrasts are paired within one frozen run with warm caches, so embedding batch variance cannot reorder systems within a comparison. *External:* see the single-domain limitation below.

*Single domain, single language.* The corpus is Chinese regulatory text; the benchmark templates and the near-duplicate structure are shaped by the domain. Replication on a second regulatory domain is future work; the construction recipe (Sect. 4) is domain-agnostic.

*Refusal pool is small* (5 queries), so Table 8 is evidence of fragility, not a powered estimate of abstention error rates.

*Run-to-run stability.* All comparisons are paired within one frozen evaluation run with warm caches; we anchor claims on within-run paired tests rather than stability of absolute decimals across reruns.

*Audit is not a gold standard.* Traceability verifies that extracted knowledge quotes its source, not that the knowledge base is complete or domain-correct (Sect. 3.3).

*Router scope.* The title-coverage signal is cheap and language-light, but its 0.5 cutoff reflects this corpus's title conventions; the sensitivity sweep (Sect. 7) is the calibration recipe for a new domain, and neural-sparse or late-interaction retrievers (SPLADE, ColBERT) were not evaluated—the comparison targets the configurations actually deployed.

## 9 Conclusion

On a small, near-duplicate regulatory corpus—the regime real governance deployments live in—no retrieval configuration wins across query types: plain BM25 owns title lookup, graph-expanded lexical retrieval owns entity and multi-hop queries, and a strong dense-plus-reranker pipeline is significantly worst on every type. A deterministic two-system router—title coverage to BM25, everything else to the graph—is statistically indistinguishable from the per-type oracle and beats every single system. No configuration's scores separate answerable from refusal queries. The practical recipe our deployment adopted is: keep the ontology evidence-audited, put BM25-honest title matching behind a one-signal router, and abstain on evidence coverage rather than retrieval score. We release everything needed to re-run this study on a new domain.

**Reproducibility Statement.** The frozen run artifact, benchmark, ablation, router, audit, and significance scripts are in the anonymized repository (footnote 1). Every model call is cached; all numbers regenerate deterministically from the frozen snapshot. During preparation we found and fixed a benchmark-builder bug that emitted non-unique question ids (306 items mapping to only 22 ids), silently corrupting id-keyed per-item analyses; all numbers in this paper come from the corrected benchmark, and the pre-fix artifacts are retained alongside for provenance.

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